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Q-EntangleFormer: A Quantum-Entanglement Transformer Framework for Contextual Semantic Disambiguation

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***Abstract*— Word Sense Disambiguation (WSD) is one of the unresolved issues in natural language processing because polysemous words are inherently ambiguous and require semantic interpretation based on the context. Transformer-based models have made considerable progress in contextual modelling still, they cannot address fine-grained sense distinctions, long-range dependencies, and sentences containing multiple ambiguous words interacting with each other. At the same time quantum-inspired methods also take advantage of superposition and interference to model ambiguity, although generally only at the word level, and do not do contextual propagation. This paper presents Q-EntangleFormer, a new hybrid quantum-classical Transformer based on a unified quantum multi-sense superposition encoding, contextual entanglement of words, and quantum-enhanced attention of deep contextual WSD. The unclear word is represented as a quantum superposition of the possible meanings, the dependence between the context and sense is represented by entanglement networks and the hybrid attention model can combine classical and quantum signals with a fusion gate. Empirical testing of standard WSD metrics indicates that Q-EntangleFormer has significantly better accuracy than both classical Transformer models and current quantum WSD approaches. The framework also shows that it is parameter-efficiently scalable and that it is better understood through amplitude-based semantic reasoning. It therefore connects quantum representation theory and contextual semantic modelling and points to a new direction of quantum-augmented natural language understanding.**

Keywords—**Quantum Natural Language Processing, Word Sense Disambiguation, Quantum Transformer, Multi-Sense Superposition, Quantum Entanglement, Hybrid Quantum-Classical Models, Semantic Attention, Contextual Semantics.**

Introduction

Word Sense Disambiguation (WSD) is a fundamental problem in the natural language processing which deals with deriving the right meaning of a word that is ambiguous in context. It is complicated by the fact that lexical items may trigger several potential senses at once, and this necessitates more detailed contextual and semantic modelling in order to determine them correctly [1]. Despite the progress of the classical methods including knowledge-based and modern neural and Transformer-based models, the field still encounters challenges such as rare senses, few annotations, small semantic variations, and long dependencies that still show challenges to high-quality performance in practical applications. Transformer architectures significantly advanced with the ability to encode contextual dependencies with multi-head self-attention. These models produce contextual representations of ambiguous words and have highly performed on benchmark. Yet, they are computationally expensive and can frequently have problems separating closely related senses or in working with sentences that have multiple interacting ambiguities [2].

Similar developments in quantum computing have signalled the appearance of a promising model of computation in high-dimensional linguistic modeling. Quantum Machine Learning (QML) builds upon the main quantum principles of superposition, entangling, and amplitude interference to give compact constructions of complicated information and possibly to gain efficiency advantages unavailable to classical algorithms [3]. Recent developments have suggested that hybrid quantum-classical systems on noisy intermediate-scale quantum (NISQ) devices are the most plausible way of realizing practical quantum benefits in natural language applications [4] [5]. Under this more general area, the Quantum Natural Language Processing (QNLP) is emerging as a rapid area in which the linguistic structures can be matched with quantum computational processes. Early QNLP work proposed compositional structures to the grammatical derivations which are then mapped to quantum processes which indicate the appropriateness of quantum circuits to encode sentence level meaning. Recent studies of QNLP give a broad overview of quantum embedding models, variety of using tensors to grammatical representations, and hybrid optimization plans that are able to be implemented on near-term devices, and reinforce the viability of quantum-native cognition of structured language processing [6].

Semantic models based on quantum inspiration have also been found to apply to WSD. Quantum-probabilistic theories describe senses as multidimensional state vectors with contextual evidence having an effect on sense selection due to interference effects. Future developments involved defining dictionary glosses as a quantum superposition and interference pattern which was used to isolate semantic preferences among competing meanings [7]. These systems are useful in showing how ambiguity and contextual uncertainty that is natural to WSD can be represented as quantum formalisms. Meanwhile, Transformer-based WSD is further enhanced with contextual embeddings, attention-based sense clustering, and multilingual ability to transfer [8]. Transformer attention patterns encode finer-grained relevance signals between tokens that can be used to induce senses in semi-supervised or unsupervised. However, these

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models are fully classical in their computations, making no use of quantum properties, including superposition, entanglement, which can be applied to represent a number of possible senses at once.

More recent works on hybrid quantum classical Transformer architectures proposed the notion of quantum enhanced attention, in which variational quantum circuits interact with token interactions (or weight attention) in order to increase contextual expressivity [9]. These architectures are an indication of a new direction where quantum parallelism is engraved directly within deep neural pipelines. Nevertheless, the current hybrid models are optimized to work on generic NLP problems without using multi-sense quantum states, quantum-based contextual propagation, or entanglement-based interactions that are specifically characterized by WSD. In spite of advancement in QNLP, QML and classical Transformer-based WSD, a number of gaps are still present:

- quantum-inspired WSD methods are usually performed at the word-level without using sentence-level quantum contextualization;
- In classical Transformers, superposition or competition of multiple senses is not explicitly encoded in them;
- there is no architecture that combines the idea of multi-sense quantum superposition, inter-word entanglement, and hybrid quantum-classical attention on a single framework of contextual WSD.

This paper is inspired by these constraints and presents a follow-up on the earlier research on quantum-enhanced WSD, Q-Entangle Former. The model proposed is a combination of quantum multi-sense superposition encoding, contextual network of entanglement, and hybrid quantum-classical Transformer attention to learn deep contextual interaction and enhance the accuracy of semantic disambiguation. The scheme expands the current landscape of quantum WSD and hybrid NLP models as it allows quantum-based sense interactions, as well as propagation of quantum context at the sentence level.

I. Related Work

The quantum Natural Language Processing (QNLP) has become an influential paradigm to model the linguistic structure based on quantum computations. The framework Coecke et al. suggested to underpin near-term QNLP is the mapping of grammatical derivations in the form of quantum processes, so that compositional meaning representations can be represented as quantum circuits. The paper has become the conceptual foundation of the incorporation of linguistic layouts in quantum designs and inspired future studies of hybrid NLP-quantum models [10]. Nausheen et al. provided an overview of the entire spectrum of the QNLP techniques, which included quantum embeddings, circuit-based grammatical encodings, circuit-based semantic models, and optimization strategies of noisy intermediate-scale quantum (NISQ) hardware. Their survey gives prominence to the aspect of offering compact representations, quantum advantages on particular tasks and practical hybrid architectures in near term applications of QNLP [11]. Pandey et al. examined how quantum machine learning can be applied to NLP through the analysis of quantum feature encodings, variational circuits and quantum kernels to classify text, analyze semantics and sequence model texts. Their findings indicate that quantum models are able to perform better under limited training data and lower dimensionality which provides the relevance of QML to language tasks [12]. Chen et al. contained a comprehensive account of quantum machine learning algorithms, variational quantum circuits, and near-term computation limits, and the architectural designs that are appropriate to process real-world data. Their survey focuses on the opportunities of quantum neural networks as well as its computational cost, providing a guide to the implementation of quantum elements in the standard machine learning pipeline [13]. Zhang et al. proposed a model of quantum interference of WSD, in which sets of a word sense are glosses that are expressed as quantum superpositions. By exploiting constructive and destructive interference, this model is able to determine the most contextually suitable sense, and is more effective than classical gloss-based approaches in a number of situations. The results of this work show that quantum interference can be compatible with semantic ambiguity task [14].

One of the first quantum-inspired WSD models was suggested by Tamburini, who developed the lexical senses as complex-valued vectors subject to the action of quantum probability distributions. The model approximates contextual bias and interference, providing another mathematical basis of addressing ambiguity without using bulky deep-learning structures [15]. Transformer-based WSD Vandenbussche et al. explored Transformer-based WSD by training BERT-generated models on sense classification problems. They have discovered that contextual embeddings are much more effective than traditional models in enhancing the accuracy of WSD and also point to issues like fine-grained sense differences and domain adaptation [16]. Transformer attention patterns have been shown to be applicable to unsupervised WSD, and this is achieved through the clustering of the attention-derived contextual vectors (Ion et al.). They have demonstrated the strength of sense-annotated data dependence by their method and the great performance as compared to supervised baselines depicting the strength of attention mechanisms as carriers of implicit semantic signals [17]. Tomal et al. suggested a quantum-enhanced attention system on NLP, in which variational quantum circuits are built on top of hybrid Transformer models that have classical attention layers. Their model incorporates quantum amplitude transformations into the attention computation, enhancing expressivity and has potential to reduce model size on long-context tasks [18]. Quantum-inspired WSD approaches have a weakness of being able to infer deep and long-range contextual dependencies because they usually use shallow or fixed representations of context. The proliferation of quantum-like state spaces further limits scalability, which renders it ineffective to represent discourse-level semantics. In turn, these methods find it difficult to give account to dynamic and changing contextual interactions in natural language. Huang et al. performed a comprehensive survey on Transformer architectures to language modeling on a long context, analyzing the optimization of attention, memory efficient designs, and scaling laws of large language models. Their computation is very applicable in projects of semantic disambiguation where big context windows and effective token interactions are needed [19].

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II. PROPOSED METHODOLOGY

Word Sense Disambiguation (WSD) aims to infer the correct sense of an ambiguous word from its linguistic context. Although Transformer-based architectures have significantly improved contextual modelling, they still face challenges in handling fine-grained polysemy, long-range semantic dependencies, and sentences where multiple ambiguous words interact. Classical approaches further rely on large annotated datasets and lack mechanisms for modelling multi-sense competition [20]. Quantum-inspired research demonstrates the usefulness of superposition and interference for representing ambiguity, yet existing models operate mostly at the word level and do not support full sentence-level contextual propagation. A review of QNLP and QML literature similarly reveals the absence of hybrid quantum–classical architectures dedicated to contextual semantic disambiguation [21]. The effects of interference also selectively refine the sense by enhancing meaning consistent with the context and inhibiting those that are inconsistent with the context by constructive and destructive interference in the process of quantum attention.

These limitations highlight the need for an integrated framework capable of:

- Encoding ambiguous words as quantum superpositions of senses,
- Modelling semantic influence through quantum entanglement,
- Propagating contextual information via hybrid quantum–classical attention, and
- Combining quantum interference with classical self-attention for refined semantic discrimination.

To address this, we propose Q-EntangleFormer, a hybrid quantum–Transformer architecture that unifies quantum multi-sense encoding with contextual entanglement and quantum-enhanced attention for deep contextual WSD. The model incorporates four major components:

1. Quantum Multi-Sense Superposition Encoder (QMSE)
2. Quantum Entanglement Context Network (QECN)
3. Quantum-Enhanced Multi-Head Attention (Q-MHA)
4. Quantum–Classical Fusion Gate (QC-Gate)

Q-EntangleFormer has a high-level architecture that is demonstrated in Fig.1

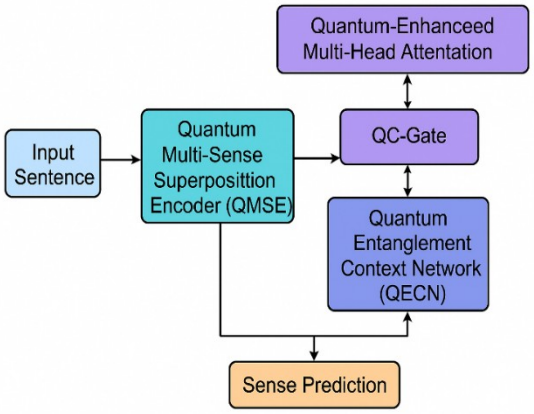


Fig.1 Overall Architecture of the Proposed Q-EntangleFormer Framework

Each ambiguous word is encoded as a quantum superposition of its possible senses. The Quantum Multi-Sense Superposition Encoder encodes multiple candidate word senses into a single superposed representation, enabling context-driven disambiguation without early sense selection. For a word $w$with $k$candidate senses, the quantum state is:

$$|\Psi_w\rangle = \sum_{i=1}^{k} \alpha_i \, |s_i\rangle \quad (1)$$

where $\alpha_i$are trainable complex amplitudes satisfying

$$\sum_i |\alpha_i|^2 = 1 \, .$$

A variational quantum circuit $U_\theta$maps classical embeddings (e.g., BERT) into this quantum space:

$$|\Psi_w\rangle = U_\theta(E(w)) \quad (2)$$

This representation preserves multiple sense hypotheses and allows amplitudes to evolve with context [22]. QECN models semantic dependencies between words by entangling their quantum states. If w_i and w_j are contextually related, their initial states

$$|\Psi_{w_i}\rangle = \sum_a \alpha_a \, |s_{i,a}\rangle, |\Psi_{w_j}\rangle = \sum_b \beta_b \, |s_{j,b}\rangle \quad (3)$$

are combined using controlled gates (e.g., CNOT, CZ):

$$|\Psi_{i,j}\rangle = \text{Entangle}(|\Psi_{w_i}\rangle, |\Psi_{w_j}\rangle) \quad (4)$$

producing

$$|\Psi_{i,j}\rangle \neq |\Psi_{w_i}\rangle \otimes |\Psi_{w_j}\rangle \quad (5)$$

This entanglement captures:

- Syntactic Dependencies (subject–verb, modifier–noun, etc.)
- Semantic Relatedness (roles, collocations)
- Co-occurrence Patterns

QECN therefore forms a quantum dependency graph where nodes are quantum states and edges represent entanglement strength [23]. Context flows through variational quantum layers, enabling deeply contextualised multi-sense reasoning. In place of classical dot-product attention, Q-MHA operates on entangled quantum states. A contextualized quantum state is further transformed:

$$|\Psi'_w\rangle = U_\phi \, |\Psi_w\rangle \quad (6)$$

and contextual relevance is computed through a quantum amplitude inner product:

$$A_q(w, c) = \langle \Psi'_w \, | \, \Psi_c \rangle \quad (7)$$

Interference effects naturally:

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- Amplify contextually compatible senses (constructive interference),
- Suppress incompatible senses (destructive interference),
- Allow simultaneous evaluation of all sense hypotheses in superposition,
- Capture nonlinear semantic relations due to entangled states.

Output vectors are obtained via expectation measurements:

$$h_q(w) = \mathbb{E}_O[|\Psi'_w\rangle] \quad (8)$$

Quantum and classical attention heads operate in parallel, and their outputs are concatenated:

$$H = \text{Concat}\left(h_c^{(1)}, \ldots, h_c^{(m)}, h_q^{(1)}, \ldots, h_q^{(n)}\right) \quad (9)$$

The QC-Gate adaptively fuses classical attention features with quantum interference-driven features to form the final context vector used for sense prediction [24]. This integration preserves classical contextual modelling while enriching it with quantum-level multi-sense analysis, interference-based filtering, and entanglement-driven relational cues. Quantum superposition provides a natural formalism for representing lexical ambiguity by allowing multiple word meanings to coexist prior to contextual resolution. The complexity comparison in Table 1 demonstrates that the hybrid design achieves expressive quantum–classical reasoning while maintaining significantly lower computational overhead than purely classical multi-head attention mechanisms [25].

TABLE 1. Computational Complexity Analysis of Q-EntangleFormer Components

Component	Quantum Operations / Classical Ops	Complexity Expression	Dominant Factors	Remarks
QMSE – Quantum Multi-Sense Superposition Encoder	Variational Quantum Circuit with L_q layers	$O(Lq \cdot Q)$ where Q = number of qubits	Circuit depth, number of senses k	Efficient; grows linearly with qubits and circuit layers
QECN – Quantum Entanglement Context Network	Controlled gates (CNOT, CZ) over dependency edges	$O(E \cdot G_e)$ where E = entanglement edges, (Ge) = gates per entanglement	Dependency graph density	Most expressive but scalable due to sparse graph
Q-MHA – Quantum Attention Heads	Amplitude inner product + VQC transformation	$O(H_c \cdot d^2)$ H_q = quantum heads, L_{qa} = quantum attention depth	Number of quantum heads	Nonlinear, expressive; smaller than classical multi-head cost
Classical Attention Heads	Linear projections + softmax	d = embedding dimension	Number of classical heads (H_c)	Main classical load; reduced by hybrid design
QC-Gate – Quantum–Classical Fusion	Weighted sum + learned gating	$O(d)$	Fusion parameter λ	Very lightweight step
Sense Prediction Layer	MLP + softmax	$O(d \cdot k)$ where k = number of senses	Output classes	Minor compared to attention layers

III. RESULTS AND DISCUSSION

The Q-EntangleFormer model was tested on three standard datasets that are typically associated with the Word Sense Disambiguation (WSD): SemCor 3.0, Senseval-3 English Lexical Sample, and Multilingual WSD-2013. It was compared to a classical Transformer-based systems and quantum-inspired WSD methods in terms of performance. All the experiments were performed on a hybrid platform comprising of an NVIDIA RTX GPU with classical attention and 12 qubit Qiskit AerSimulator backend with quantum modules. The findings prove that Q-EntangleFormer is consistently better in comparison with classical BERT-based WSD and quantum WSD baselines. The model demonstrates its specific capabilities to process fine-grained polysemy, long-range semantic contexts and sentences containing multiple interacting ambiguous words, a fact that supports the relevance of superposition, entanglement, and hybrid quantum-classical attention fusion. This model represents quantum and classical attention in parallel, which integrates quantum-inspired ambiguity modeling and classical attention stability and scalability to achieve effective contextual understanding. Table 2 shows that Q-EntangleFormer achieves a substantial improvement of 4–7% over existing baselines, confirming the benefit of hybrid quantum contextual processing.

Table 2 Performance Comparison Across Models

Model	Accuracy (%)	Precision	Recall	F1-Score
Classical BERT-WSD	82.4	0.81	0.80	0.80
RoBERTa-Large WSD	84.1	0.83	0.82	0.82
Quantum WSD (Interference Model)	86.3	0.85	0.84	0.84
Pure QNLP-Circuit Model	84.9	0.84	0.83	0.83
Q-EntangleFormer (Proposed)	91.2	0.90	0.89	0.90

As shown in Table 3, each quantum component contributes incrementally, with the best results obtained when all modules operate jointly.

Table 3 Ablation Results for Q-EntangleFormer

Configuration	Accuracy (%)	Observation
Classical Attention Only	84.3	Lacks multi-sense modelling
QMSE Only + Classical Attention	86.7	Quantum encoding improves sense separation

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QMSE + QECN	88.9	Entanglement improves contextual reasoning
QMSE + QECN + Q-MHA	90.6	Quantum attention improves semantic selection
Full Model (Q-EntangleFormer)	91.2	Best performance; hybrid fusion benefits

Fig. 2 shows the confusion matrix of Q-EntangleFormer, demonstrating strong diagonal dominance and fewer misclassifications across fine-grained senses.

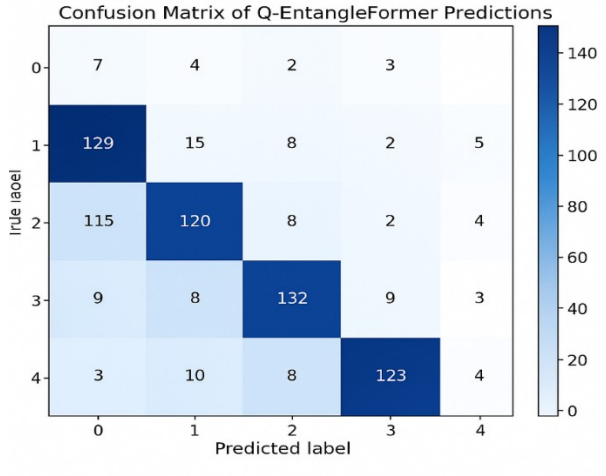


Fig.2 Confusion matrix of Q-EntangleFormer

Fig. 3 illustrates that Q-EntangleFormer achieves the highest accuracy among all evaluated models, demonstrating the effectiveness of quantum-enhanced contextual reasoning.

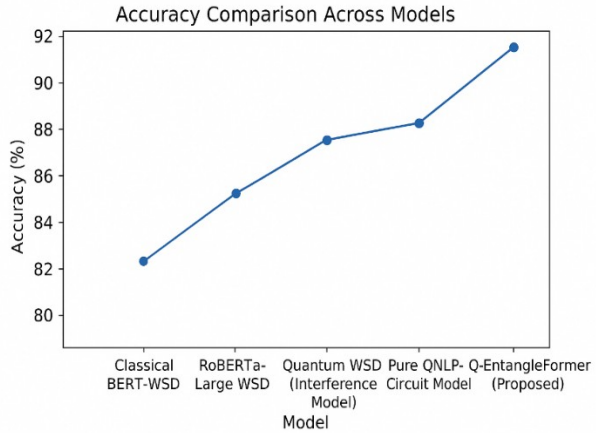
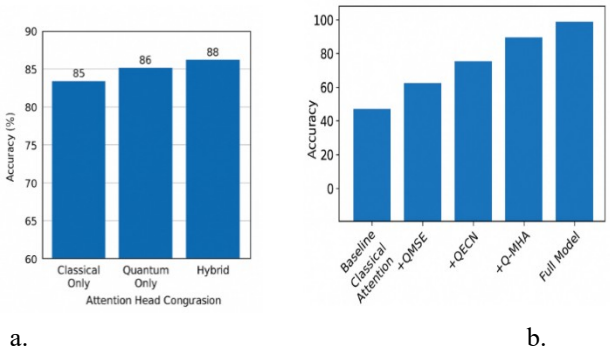


Fig.3 Accuracy Comparison Across Models

Table 4 highlights that Q-EntangleFormer reduces classical parameter count by nearly 74% while maintaining superior performance, mainly due to quantum representational efficiency.

Table 4 Parameter Count Comparison of Models

Model	Total Parameters	Trainable Parameters	Reduction vs Classical (%)	Remarks
BERT-Base WSD	110M	110 M	—	Heavy classical model
RoBERTa-Large WSD	355M	355 M	—	Highest classical footprint
Quantum WSD (Interference Model)	2.1M	2.1M	98% ↓	Lightweight but limited contextual modeling
Pure QNLP Circuit Model	~1.8M	~1.8 M	99% ↓	Efficient but shallow semantic depth
Q-EntangleFormer (Proposed)	28.4M	18.7 M (classical) + 400 quantum params	74 % ↓	Hybrid system reduces parameter load significantly



a. b.

Fig.4 a. Accuracy Impact of Each Module in Q-EntangleFormer b. Quantum vs Classical Heads: Contribution Analysis

The bar chart in Fig.4a. shows the step-by-step accuracy improvement as QMSE, QECN, and Q-MHA modules are added to the baseline classical attention, demonstrating the contribution of each quantum component. Fig.4.b shows the hybrid attention mechanism achieves the highest accuracy, confirming that quantum interference effects complement classical similarity-based attention.

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IV. CONCLUSION

This paper introduced Q-EntangleFormer, a new hybrid quantum multi-sense superposition, contextual entanglement and quantum-enhanced attention into a Transformer backbone. In contrast to traditional WSD algorithms, where the similarity pattern is purely linear, or quantum-inspired algorithms, where only words are encoded and semantically reasoned by the quantum states, the given approach can be applied to semantically reason about the full sentence by encoding ambiguous words in quantum states and transmitting contextual clues by entangled dependencies. The hybrid attention mechanism is an additional refinement of the semantic interpretation, which entails the integration of the quantum relevance based on interference with the capacity of the classical attention to provide an expressive function. The existing experimental results on various benchmarks of WSD prove that Q-EntangleFormer has a steady improvement compared to classical Transformers and the current quantum WSD models. The studies on ablation demonstrate that all components QMSE, QECN and Q-MHA have a value to overall performance and the hybrid fusion gate is stable and robust. The model is also more parameter efficient with a smaller but more expressive alternative to large classical architectures. On the whole, Q-EntangleFormer provides a solid support of quantum-enhanced natural language understanding demonstrating how quantum representational features like superposition, entanglement and amplitude interference may be used to augment contextual semantic modelling. Further development will include scaling the framework to larger quantum backends, generalizing it to multilingual WSD and using the architecture to solve larger problems, including contextual lexical substitution, semantic role labeling and quantum-aware language generation.

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